The question

We only ever used **100 of FoldBench's 334** monomers. #213 showed the contact eval set is 58% homologous to exp199's 70.9M training sequences, and that what survives a homology filter is mostly de novo *designs* — the **natural** count bottoms out around 50, too thin to measure novel-protein performance rather than bound it.

So: if we grow the eval set with the FoldBench monomers we never used, how many of the expanded set survive a sequence-identity filter against our training data, at 40% and at 30%?

No new inference. exp213's 17 GB MMseqs2 target database is reused as-is; this appends 222 net-new queries to its 554 and re-runs the search. About seven minutes on the workstation.

The set, verified not assumed

FoldBench's `monomer_protein.csv` at exp12's pinned commit `4273f687`, sha256-verified, has 334 rows, and our 100 are exactly its first 100 — checked rather than taken on faith. That leaves 234 unused. **Twelve of those are already eval proteins** under exp65's `denovo_pdb` label and would have been double-counted, so the real increment is **222 net-new** and the expanded set is **776**.

Sequences come from RCSB's canonical `entity_poly` field. FoldBench's `chain_id` is sometimes the mmCIF *label* asym id rather than the auth chain — ten entries, not the five the issue listed — so chains resolve auth-first with a label fallback and fail loudly on ambiguity.

Both residue checksums land exactly (66,692 aa and 64,624 aa), and all 100 of the sequences we already use are reproduced **byte-for-byte** through the new fetch path.

Parity with #213 is exact

Re-running exp213's 554 queries through the expanded search reproduces its **284 and 264** survivors at the two thresholds, and its 273 and 255 without the coverage gate, with **zero of the 554 rows** changing identity or stratum. The expanded table is a strict superset of exp213's and joins to it on dataset and stem.

One correction to the plan on the issue: its quoted command line says `--max-seqs 2000`, but exp213's published table was built with **5000** — 2000 was its AFDB-only cross-check. Parity means matching the published table.

Result — 23 survivors, not the 33 predicted

Of the 222 net-new proteins, **23 survive a <40% identity filter and 11 survive <30%**, taking the expanded eval set to 307 of 776 and 275 of 776.

The extrapolation on the issue was optimistic at both thresholds, which is the question it could not answer for itself. It predicted 33 and 24. The 100 monomers we already use are the *oldest-deposited* rows of a PDB-ID-sorted file, and the newer entries turn out to be **more** homologous to our training data, not less: 10.4% survive at <40% against 15.0%, and 5.0% at <30% against 11.0%. Neither gap clears $p<0.05$ alone (Fisher 0.26 and 0.057), so this reads as a consistent shortfall rather than a proven rate difference — but the counts the eval set actually gets are 23 and 11.

Length is not the explanation: median 242 aa against 247.5 aa.

The natural count is what decides this

Every one of the 23 new survivors is a **natural** protein. The decontaminated natural count goes from **55 to 78 at <40%**, a gain of 23 or **+42%**, and from **50 to 61 at <30%**, a gain of 11 or +22%.

The baseline is **55, not the 58** the issue quotes. #213 splits designed from natural on the dataset label, which cannot see a designed protein sitting in a FoldBench row — and one can, since 12 FoldBench monomers are themselves in exp65's de novo set. Resolving each FoldBench entity's RCSB source organism finds that 3 of #213's 15 FoldBench-100 survivors at <40% are synthetic constructs. The proxy is calibrated, flagging 12 of 12 known designs, and deliberately conservative, since it also catches engineered variants of natural proteins. Only 4 of the 222 trip it, and none of those 4 survive either filter.

Verdict

Fold it in at <40%; it changes little at <30%. A +42% increase in decontaminated natural proteins is a real gain on exactly the axis #213 said bottoms out. The +22% at <30% is not enough to change what the eval set can measure.

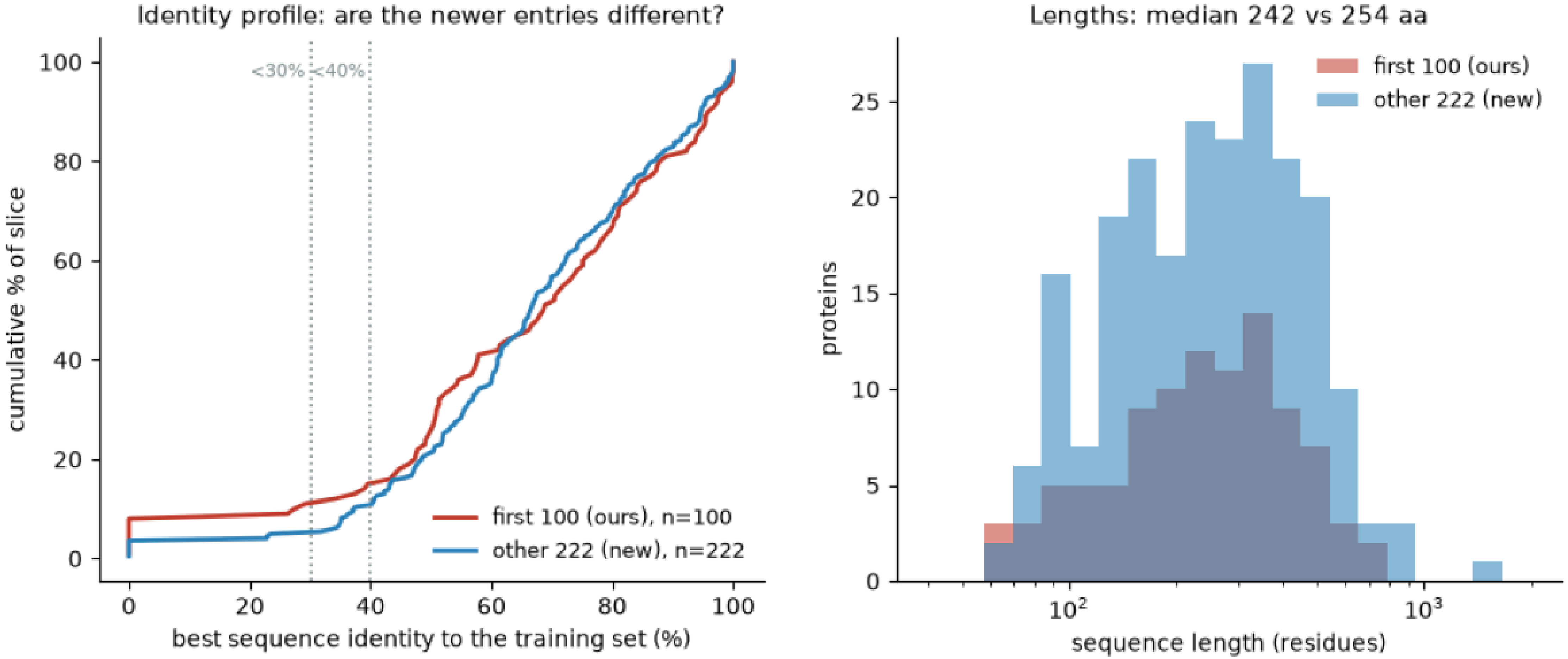
FoldBench is confirmed as the dirtiest slice we have. **89.6%** of the net-new monomers fail a 40% filter, and **all 222** have at least some alignment into the training set, so the expansion adds only **8** proteins to #213's "no detectable homolog" stratum, 231 to 239. Of the 199 dropped at <40%, 174 hit both training arms — the metagenomic half is not the marginal contaminator here.

Recommendation: add `foldbench_rest` as its own stratum rather than merging it into `foldbench100`, since the two have measurably different training-set proximity and any future "first N FoldBench rows" would inherit the same deposition-date bias.

Not measured here: the fold-novel count for the 222, which needs a Foldseek pass against exp41's Modal-hosted AFDB representative DB plus 222 structure downloads — beyond this issue's sequence-search budget.

identity_profile_old_vs_new.png

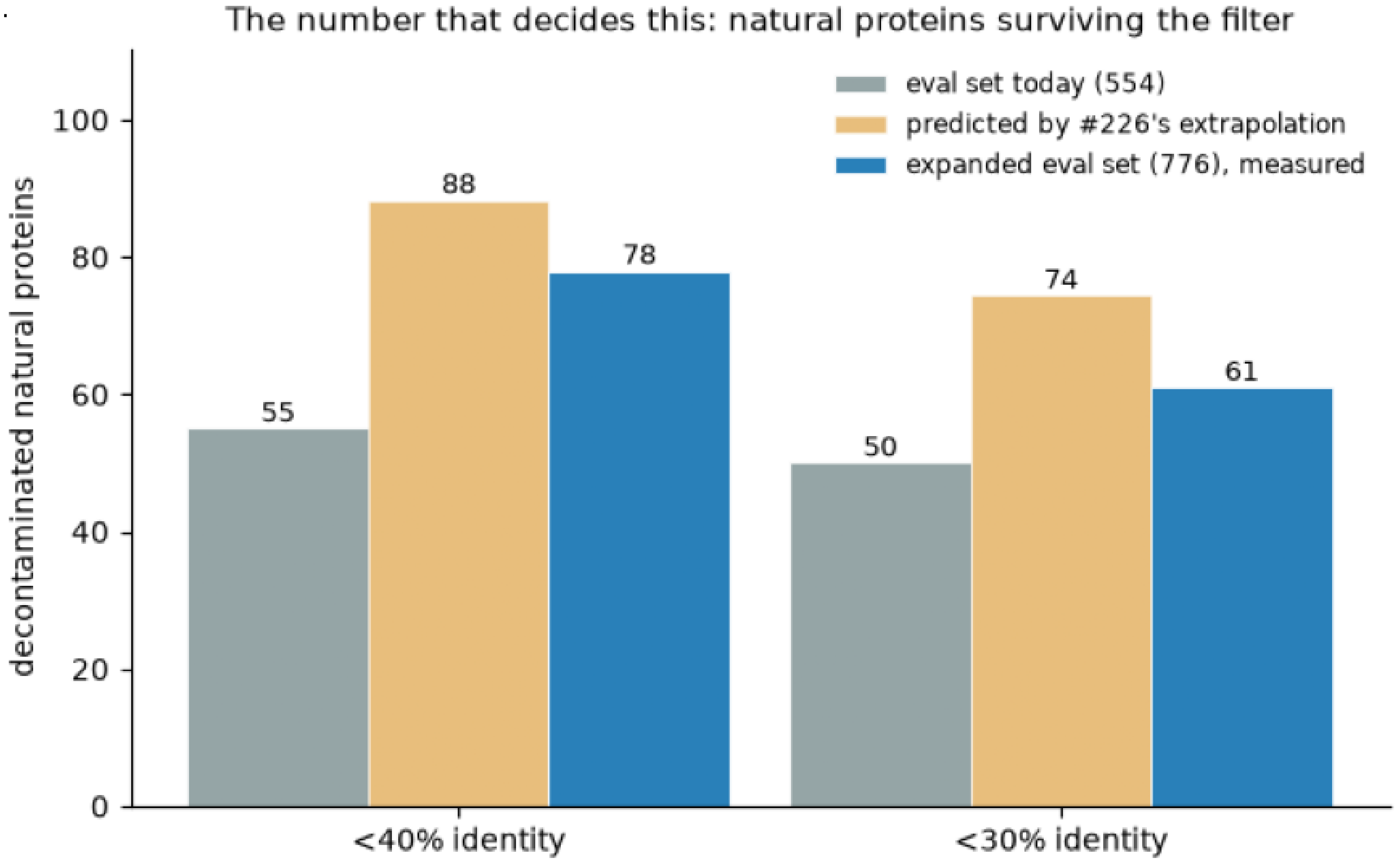
Left: cumulative distribution of best training-set identity; proteins with no covered hit are plotted at 0. The new curve sits to the right of the old one below 40 %, which is the shortfall against the extrapolation. Right: the length profiles match, so length is not the explanation.



$ plot_survival.py

natural_gain.png

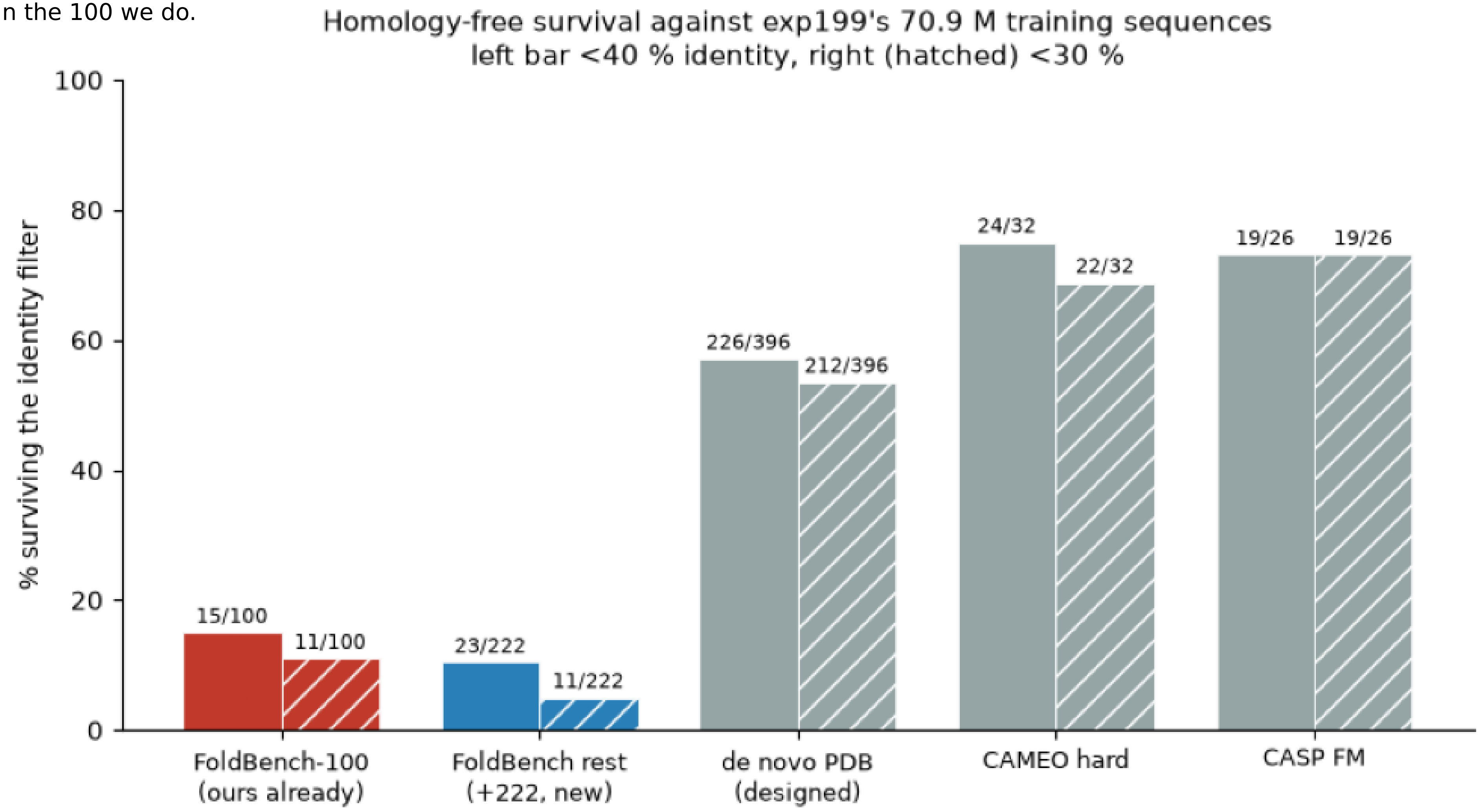
Natural (non-designed) eval proteins with no training homolog above the threshold. 'Natural' excludes both exp65's de novo set and any FoldBench entity whose RCSB source organism is synthetic. The expansion delivers its <40 % gain but only about half the predicted <30 % gain.



$ plot_survival.py

survival_by_dataset.png

Share of each eval-set slice with no training sequence above the identity threshold (MMseqs2 -s 7.5, hit counted at E<=1e-3 and qcov>=0.50). Bar labels are survivors/total. The two FoldBench slices are the dirtiest in the eval set; the 222 we never used are dirtier still than the 100 we do.



$ plot_survival.py